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Secure Computation Technologies

Lecture Handout

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Privacy Enhancing Technologies (PETs)

Privacy Enhancing Technologies (PETs), offer a way for companies to process data, whilst ensuring the privacy or confidentiality of the data remains in place, and at the same time allowing sharing data with external partners, or processing sensitive data. PETs allow online users to protect the privacy of their personally identifiable information (PII), which is often provided to and handled by services or applications. PETs use techniques to minimize an information system's possession of personal data without losing functionality.

1. Motivation for Secure Computation

Fundamental (Secure Computation) research question:

“How to allow the collection and purposeful processing of private data, without compromising individual privacy?”

2. Secure Computation Technologies at a glance

2.1 Data Anonymization

Data anonymization is the process of protecting private or sensitive information by erasing or encrypting identifiers that connect an individual to stored data. Unfortunately, data does anonymization does not work in practice, especially when the attacker can combine information from several sources.

Latanya Sweeney did extensive research on reidentification, showing that anonymized data, is not really anonymous, with the following prime example:

The Massachusetts Group Insurance Commission had a bright idea back in the mid-1990s—it decided to release “anonymized” data on state employees that showed every single hospital visit. The goal was to help researchers, and the state spent time removing all obvious identifiers such as name, address, and Social Security number. But a graduate student in computer science saw a chance to make a point about the limits of anonymization.

Latanya Sweeney requested a copy of the data and went to work on her “reidentification” quest. It did not prove difficult. Law professor Paul Ohm describes Sweeney’s work:

At the time GIC released the data, William Weld, then Governor of Massachusetts, assured the public that GIC had protected patient privacy by deleting identifiers. In response, then-graduate student Sweeney started hunting for the Governor’s hospital records in the GIC data. She knew that Governor Weld resided in Cambridge, Massachusetts, a city of 54,000 residents and seven ZIP codes. For twenty dollars, she purchased the complete voter rolls from the city of Cambridge, a database containing,

among other things, the name, address, ZIP code, birth date, and sex of every voter. By combining this data with the GIC records, Sweeney found Governor Weld with ease. Only six people in Cambridge shared his birth date, only three of them men, and of them, only he lived in his ZIP code. In a theatrical flourish, Dr. Sweeney sent the Governor's health records (which included diagnoses and prescriptions) to his office.

This was only an early mile marker in Sweeney's career; in 2000, she showed that 87 percent of all Americans could be uniquely identified using only three bits of information: ZIP code, birthdate, and sex.

2.2 Differential Privacy

Differential privacy (DP) is a **mathematical framework for ensuring the privacy of individuals in datasets**. It can provide a strong guarantee of privacy by allowing data to be analyzed without revealing sensitive information about any individual in the dataset.

Several mechanisms are commonly used in differential privacy to ensure the privacy of individuals in datasets. One of the most commonly used mechanisms to answer numerical questions is the **addition of calibrated noise**: adding enough noise to the output to mask the contribution of any possible individual in the data while still preserving the overall accuracy of the analysis. One concrete example of adding noise for differential privacy is the **Laplace mechanism**.

Differential privacy incurs an inherent privacy leakage, which is called **Privacy Budget** in DP terms. Moreover, it comes with a tradeoff between data privacy and data utility (i.e., as privacy increases, data utility decreases, and the other way around).

2.3 Cryptographic Techniques

- **Homomorphic Encryption**

Homomorphic encryption (HE) is the conversion of data into ciphertext that can be analyzed and worked with as if it were still in its original form. Homomorphic encryption enables complex mathematical operations to be performed on encrypted data without compromising the encryption.

Homomorphic encryption allows outsourcing the computation to **one single party** (server), and we inherently need to **trust** the server to perform the computation correctly and honestly. It entails **one round** of communication (Query – Response), and it has **communication and computational cost** considerably larger than that of plaintext computations.

There are three types of homomorphic encryption schemes:

- **Partially Homomorphic Encryption**: allows one type of operations (addition or multiplication) to be performed on the ciphertexts, while allowing unlimited number of operations.

- **Somewhat Homomorphic Encryption:** allows two types of operations (addition AND multiplication) to be performed on the ciphertexts, while allowing only limited number of operations.
- **Fully Homomorphic Encryption:** allows two types of operations (addition AND multiplication) to be performed on the ciphertexts, while also allowing unlimited number of operations.
- **Multiparty Computation**

Multiparty Computation (MPC) allows a set of mutually distrusting parties to compute a function over their private inputs, without disclosing any information about those inputs (to the other computing parties, or external parties), beyond what can be inferred from the output of the computation alone.

MPC requires a *minimum of two parties* (servers) to perform a computation. *Trust* is distributed and there are various security models one can consider. *Multiple rounds* of communication are required in an interactive protocol for MPC protocols to be completed. The *communication and computational cost* is considerably larger than of plaintext computations (yet *more efficient* than HE solutions!)

3. A closer look at MPC

MPC operates under different security models. Several such models, encompassing the security assumptions the protocol designers need to make, exist, with the most notable ones being:

- The **active security** model (assuming so-called malicious adversaries), where we assume that the adversary can arbitrarily deviate from the protocol description.
- The **passive security** model (assuming a so-called honest-but-curious adversary), where we assume that the adversary tries to learn as much private information as possible, but it correctly (honestly) follows the protocol description.

Furthermore, MPC admits different corruption thresholds, indicating the capabilities of the adversary, in terms of the number of computation parties said adversary is allowed to corrupt. As such we have protocols supporting:

- An **honest majority**, meaning that the adversary is allowed to corrupt strictly less than half of the protocol participants.
- A **dishonest majority**, meaning that the adversary is allowed to corrupt a subset of the parties greater than half, with the ultimate case of **full-threshold** security, where the adversary is allowed to corrupt all-but-one MPC parties, and the security of the overall protocol is still guaranteed.

When designing MPC protocols we also need to consider the available network resources/settings, to select the most suitable solutions.

- When working over a **LAN** (local area network), our MPC protocol should focus on **minimum data transfer**, which makes **Secret-Sharing based MPC** the most suitable solution.
- When working over a **WAN** (wide area network), our MPC protocol should focus on **minimum number of rounds**, which makes **Garbled-Circuit based MPC** the most suitable solution.

Secret Sharing based MPC is usually linear (e.g., based on additive secret sharing, or Shamir secret sharing), meaning that all linear operations can be computed locally (without interaction) on the parties shares. However, multiplications require the execution of an interactive protocol.

In **Garbled Circuit** cryptographic protocols, the whole function has to be described in advance, as a circuit (arithmetic or Boolean). In the classical case, this describes a two party computation, where one party plays the role of the **garbler** (and as such cryptographically encodes the function and its own inputs into the circuit), and the other party plays the role of the **evaluator** “walking through” the circuit gate-by-gate to perform the evaluation.

MPC in the **preprocessing model**, splits the MPC computation in two phases:

- A **preprocessing (aka offline) phase**, which is data independent and can be executed anytime prior to the protocol execution. This phase prepares (random correlated) material for input sharing and multiplication operations, which can be used later (in the online phase) to compute efficiently the protocol on the real inputs.
- An **online phase**, which is data dependent, and can be executed once the inputs become available. This phase consumes the preprocessed data to efficiently complete the computation on the real inputs.

4. Application Scenarios

EPIC: Efficient Private Image Classification (or: Learning from the Masters)

Outsourcing an image classification task raises privacy concerns, both from the image provider’s perspective, who wishes to keep their images confidential, and from the classification algorithm provider’s perspective, who wishes to protect the intellectual property of their classifier. We propose EPIC, an efficient private image classification system based on support vector machine (SVM) learning, secure against malicious adversaries. EPIC builds upon transfer learning techniques known from the Machine Learning (ML) literature and minimizes the load on the privacy-preserving part. Our solution is based on Multiparty Computation (MPC), it is 34 times faster than Gazelle (USENIX’18) –the state-of-the-art in private image classification– and it improves the communication cost by 50 times, with a 7% higher accuracy on CIFAR-10 dataset. For the same accuracy as Gazelle achieves on CIFAR-10, EPIC is 700 times faster and the communication cost is reduced by 500 times.

Towards Practical Privacy-Preserving Genome-Wide Association Study

The deployment of Genome-wide association studies (GWASs) requires genomic information of a large population to produce reliable results. This raises significant privacy concerns, making people hesitate to contribute their genetic information to such studies. We propose two provably secure solutions to

address this challenge: (1) a somewhat homomorphic encryption (HE) approach, and (2) a secure multiparty computation (MPC) approach. Unlike previous work, our approach does not rely on adding noise to the input data, nor does it reveal any information about the patients. Our protocols aim to prevent data breaches by calculating the χ^2 statistic in a privacy-preserving manner, without revealing any information other than whether the statistic is significant or not. Specifically, our protocols compute the χ^2 statistic, but only return a yes/no answer, indicating significance. By not revealing the statistic value itself but only the significance, our approach thwarts attacks exploiting statistic values. We significantly increased the efficiency of our HE protocols by introducing a new masking technique to perform the secure comparison that is necessary for determining significance. We show that full-scale privacy-preserving GWAS is practical, as long as the statistics can be computed by low degree polynomials. Our implementations demonstrated that both approaches are efficient. The secure multiparty computation technique completes its execution in approximately 2 ms for data contributed by one million subjects.

References

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